Zaki Haseeb Kazi

cs25mtech11021@iith.ac.in | zakikazi178@gmail.com | +91 9112898412 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.17
Kai. Prabhakar Bhasme Uchha Madhyamik Vidyalay, Dharashiv	Class XII (Maharashtra State Board)	2019	63.08%
SSA High School, Solapur	Class X (Maharashtra State Board)	2017	89.40%

POSITIONS OF RESPONSIBILITY

Teaching Assistant | IIT Hyderabad

- Administered course logistics for three advanced technical courses: **Data Mining, Database Management Systems II, and Systems for Machine Learning**, supporting academic delivery.
- Evaluated technical assignments and comprehensive examinations, consistently applying strict academic rubrics to ensure fair and standardized grading.

SKILLS

Languages: Python, C++, C, PHP, CUDA

DevOps & Infrastructure: Linux, Docker, Kubernetes (K8s), Git, GitHub, Wireshark

RELEVANT COURSES

Systems & Networking: Advanced Computer Networks, Network Security, GPU Programming, Distributed AI Training

PROJECTS

CUDA Kernel Optimization of BANG (Billion-Scale ANN on GPU) | Group Course Project| *GitHub*

- Profiled and **optimized CUDA kernels** in BANG, an open-source billion-scale approximate nearest neighbor search library, reducing per-query latency from **21ms to 14ms (~33% improvement)**
- Identified **memory-bound bottlenecks** in kernel execution and restructured **memory access patterns** to improve throughput and reduce stalls
- Replaced merge sort with a **bitonic sort** implementation for GPU-parallel sorting, better exploiting **warp-level parallelism** and reducing **divergence**

-DPP Active Learning for Data-Efficient Multimodal Fusion | Course Project | *GitHub*

- Extended **FuseMix (CVPR 2024)** by designing a dynamic, diversity-aware **active learning curriculum** that replaces uniform data sampling with **task-aware DPP-based subset selection** for training multimodal fusion adapters
- Built a task-aware relevance signal via **EMA-tracked contrastive loss** and a **temperature-annealed softmax**
- Outperformed the reproduced FuseMix baseline across all retrieval metrics on Flickr30k and MS-COCO, improving **Image-to-Text R@1 from 47.3% to 56.0%** on Flickr30k and **35.4% to 38.6%** on COCO

TransactiWar: Secure Money Transfer Platform | Group Course Project | *Github*

- Designed and hardened a peer-to-peer money transfer web app, implementing **Argon2id password hashing, CSRF tokens, PDO prepared statements**, and **strict session management** (secure/SameSite cookies, idle/absolute timeouts) to protect accounts and transactions
- Secured user-facing input fields against **XSS and injection** via sanitization and a **strict CSP**, and enforced file-type, size, and content validation on profile image uploads to prevent malicious file execution
- Architected a **segmented Docker network** isolating the internet-facing **Nginx** layer, app server, and **PostgreSQL**

database into separate tiers, paired with a **hardened reverse proxy (TLS 1.3, HSTS, rate limiting)** to minimize attack surface
- Withstood a week-long **adversarial red-teaming exercise** where every student's app was publicly attacked by classmates for points, with **zero successful breaches** and no points lost

EXTRACURRICULAR

Batch Merch Design: Designed the hoodie for the CSE 2025 batch, which won a student-wide poll to become the year's official batch merchandise